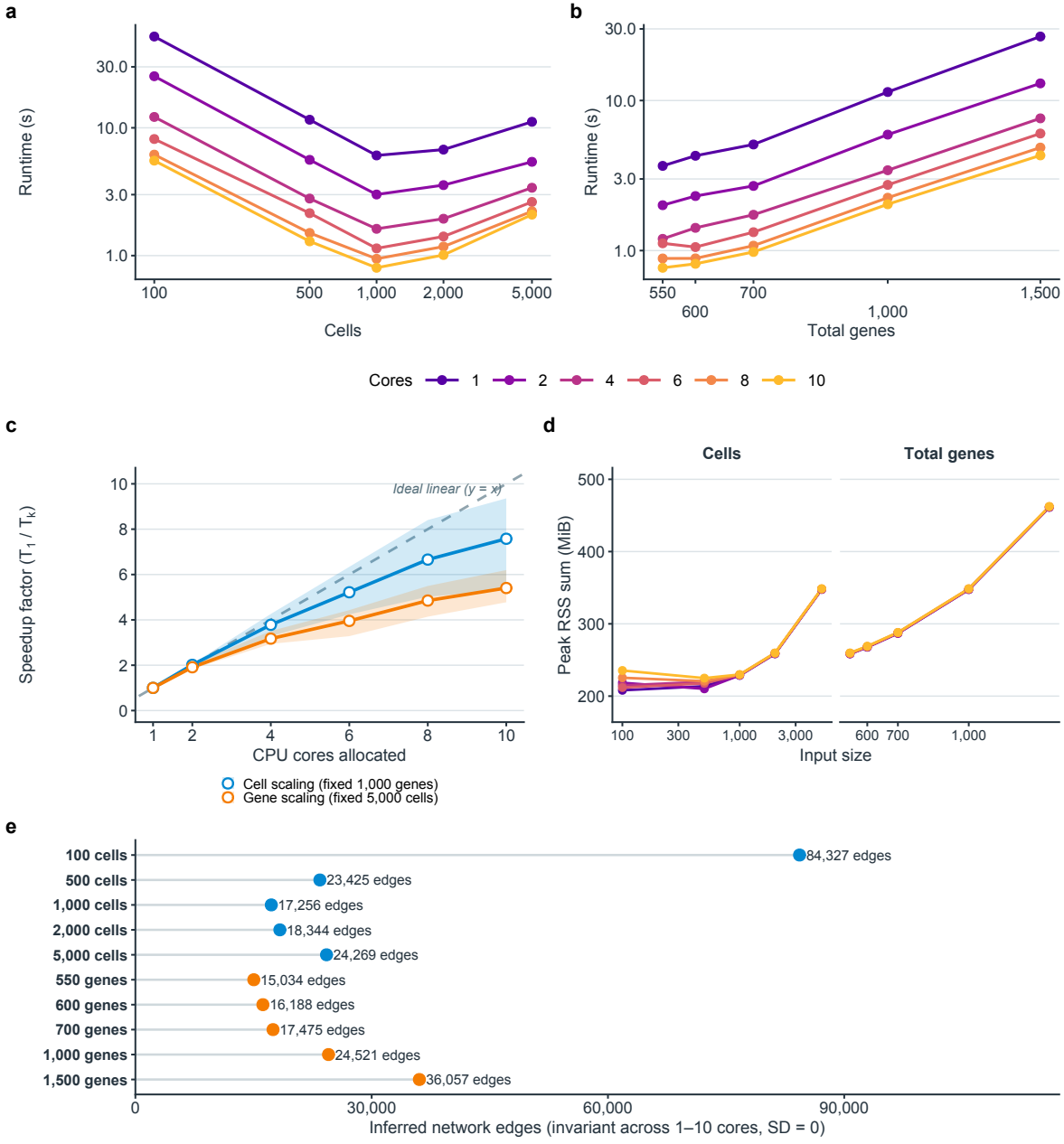
a
Runtime (s)
30.0
10.0
3.0
1.0
100
500
1,000
2,000
5,000
Cells
b
Runtime (s)
30.0
10.0
3.0
1.0
550
600
700
1,000
1,500
Total genes
Cores
1
2
4
6
8
10
c
Speedup factor (T1 / Tk)
Ideal linear (y = x)
0
2
4
6
8
10
1
2
4
6
8
10
CPU cores allocated
Cell scaling (fixed 1,000 genes)
Gene scaling (fixed 5,000 cells)
d
Cells
Total genes
Peak RSS sum (MiB)
500
400
300
200
100
300
1,000
3,000
600
700
1,000
Input size
e
100 cells
84,327 edges
500 cells
23,425 edges
1,000 cells
17,256 edges
2,000 cells
18,344 edges
5,000 cells
24,269 edges
550 genes
15,034 edges
600 genes
16,188 edges
700 genes
17,475 edges
1,000 genes
24,521 edges
1,500 genes
36,057 edges
0
30,000
60,000
90,000
Inferred network edges (invariant across 1–10 cores, SD = 0)